Package 'snp'

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Title Plot standard normal distribution

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Depends ggplot2

Description Draws the standard normal curve with score scales lined up under it: z, T, percentile, sten, stanine, grade and IQ, or scales defined in a data frame, each shown as labelled tick marks or coloured bands.

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URL https://sedzinfo.github.io/snp/, https://github.com/sedzinfo/snp

BugReports https://github.com/sedzinfo/snp/issues

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`standard_normal_custom` *Plot the standard normal curve with your own score scales*

Description

Draws the standard normal curve, marked at every standard deviation with the percentage of cases in each interval, with the scales you describe in a data frame lined up under it. A scale can be a row of labelled tick marks (like T scores), a row of coloured bands (like stanines), or both.

Usage

```
standard_normal_custom(scales = standard_normal_scales_data(), size = 3)
```

Arguments

`scales`	A data frame with one row per tick mark or band, and the columns: **scale** Name of the scale, shown at the left of its row. Rows are drawn from top to bottom in the order the names first appear. **z** Position on the z axis: where the tick mark goes, or where the band starts. **label** Text shown at the tick mark or inside the band. **z_end** Optional. Where the band ends on the z axis. Leave the column out, or use `NA`, for a tick mark. **colour** Optional. Fill colour of the band. Bands without a colour are coloured from light to dark along their row. The default, `standard_normal_scales_data()`, holds the scales of `standard_normal_scales`.
`size`	Text size of all labels, in millimetres (ggplot2 text size). The default of 3 suits figures around 7 inches wide.

Value

A ggplot object. Print it to draw the chart, or save it with `ggplot2::ggsave()`. The chart keeps a fixed aspect ratio, and every scale you add makes it taller.

See Also

`standard_normal_scales_data`, `standard_normal_scales`

Examples

```
standard_normal_custom()
# your own scales: tick marks for a test score and bands for performance levels
my_scales<-rbind(data.frame(scale="SCORE",z=-3:3,z_end=NA,label=seq(20,80,10)),
           data.frame(scale="LEVEL",z=c(-4,-1,1),z_end=c(-1,1,4),label=c("Low","Average","High")))
standard_normal_custom(my_scales)
# start from the standard scales, drop the grades and add Wechsler scaled scores (mean 10, SD 3)
```

```
scales<-standard_normal_scales_data()
scales<-scales[scales$scale!="GRADE",]
scales<-rbind(scales,data.frame(scale="SCALED",z=-3:3,z_end=NA,label=seq(1,19,3),colour=NA))
standard_normal_custom(scales)
```

`standard_normal_plot` *Plot the standard normal curve with score scales*

Description

Draws the standard normal curve, marked at every standard deviation with the percentage of cases in each interval, and lines up under it the z, T, percentile, sten, stanine, grade and IQ scales, so that a score on one scale can be read off on the others.

Usage

```
standard_normal_plot(size = 5)
```

Arguments

`size`	Text size of all labels, in millimetres (ggplot2 text size). The default of 5 suits large figures; use about 3 for figures around 7 inches wide.

Details

All scales are placed on the z axis:

- **z**: standard score, mean 0 and standard deviation 1.
- **T**: $50 + 10z$.
- **Percentile**: the z values of the 1st, 5th, 10th, 25th, 50th, 75th, 90th, 95th and 99th percentiles.
- **Sten**: ten bands half a standard deviation wide, sten 1 below $z = -2$ and sten 10 above $z = 2$ (mean 5.5, standard deviation 2).
- **Stanine**: nine bands half a standard deviation wide, stanine 5 from $z = -0.25$ to $z = 0.25$ (mean 5, standard deviation 2).
- **Grade**: groups of stanines, E (1-2), D (3), C (4-6), B (7) and A (8-9).
- **IQ**: $100 + 15z$.

The coloured sten, stanine and grade bands are drawn as thick lines, whose thickness does not follow the size of the figure, so in figures of ordinary size they spill over the rows above them. `standard_normal_scales` draws the same chart so that it fits any figure size.

Value

A ggplot object. Print it to draw the chart, or save it with `ggplot2::ggsave()`. The chart keeps a fixed aspect ratio and is about 2.5 times as wide as it is tall, so a figure with those proportions has no empty margins.

See Also

standard_normal_scales

Examples

```
standard_normal_plot()
standard_normal_plot(size=3)
```

`standard_normal_scales` *Plot the standard normal curve with score scales at any figure size*

Description

Draws the chart of `standard_normal_plot`: the standard normal curve, marked at every standard deviation with the percentage of cases in each interval, with the z, T, percentile, sten, stanine, grade and IQ scales lined up under it, so that a score on one scale can be read off on the others. Here the coloured bands are sized in plot units, so the rows never overlap whatever the size of the figure, each element is drawn once, and no ggplot2 warnings are raised.

Usage

```
standard_normal_scales(size = 3)
```

Arguments

`size`	Text size of all labels, in millimetres (ggplot2 text size). The default of 3 suits figures around 7 inches wide; for larger text, use a larger figure so that the labels keep their room.

Details

All scales are placed on the z axis:

- **z**: standard score, mean 0 and standard deviation 1.
- **T**: $50 + 10z$.
- **Percentile**: the z values of the 1st, 5th, 10th, 25th, 50th, 75th, 90th, 95th and 99th percentiles.
- **Sten**: ten bands half a standard deviation wide, sten 1 below $z = -2$ and sten 10 above $z = 2$ (mean 5.5, standard deviation 2).
- **Stanine**: nine bands half a standard deviation wide, stanine 5 from $z = -0.25$ to $z = 0.25$ (mean 5, standard deviation 2).
- **Grade**: groups of stanines, E (1-2), D (3), C (4-6), B (7) and A (8-9).
- **IQ**: $100 + 15z$.

Value

A ggplot object. Print it to draw the chart, or save it with `ggplot2::ggsave()`. The chart keeps a fixed aspect ratio and is about 2.5 times as wide as it is tall, so a figure with those proportions has no empty margins.

See Also

`standard_normal_plot`

Examples

```
standard_normal_scales()
standard_normal_scales(size=4)
# save it with the chart's proportions
ggsave(file.path(tempdir(),"standard_normal_scales.png"),standard_normal_scales(),width=7,height=2.8)
```

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standard_normal_scales_data

Score scales of the standard normal chart, as a data frame

Description

Returns the z, T, percentile, sten, stanine, grade and IQ scales of `standard_normal_scales` in the format `standard_normal_custom` takes, so you can start from them and change, remove or add scales.

Usage

```
standard_normal_scales_data()
```

Value

A data frame with the columns scale, z, z_end, label and colour, and one row per tick mark (z, T, percentile and IQ, with `z_end` and `colour` set to `NA`) or per band (sten, stanine and grade).

See Also

`standard_normal_custom`

Examples

```
scales<-standard_normal_scales_data()
head(scales)
unique(scales$scale)
```

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